

Trees save lives

But not in the way you're thinking

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Available from: <https://github.com/maxbiostat/presentations/>

Plan for today

Problem: Extracting epidemiological insight from DNA

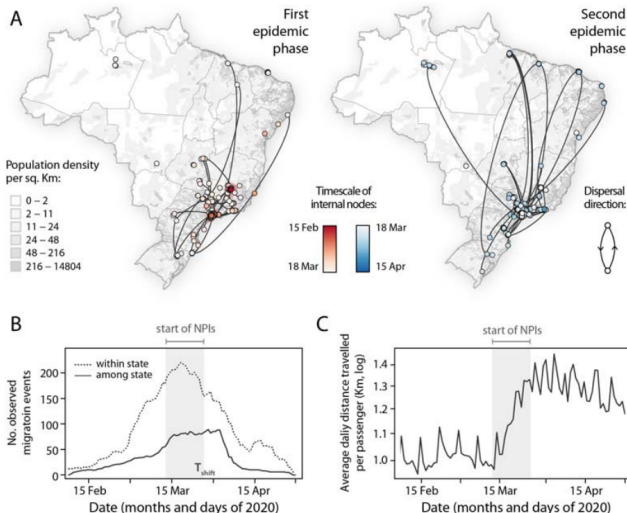
- ⊙ MCMC in tree space: a journey through a strange land;
- ⊙ Validating Bayesian evolutionary models;
- ⊙ Open problems.

Where the magic happens

Academic life at FGV EMap

Dealing with huge complex data

Where did this virus come from?¹



¹Cândido et al. (2020) *Science*.

Motivation

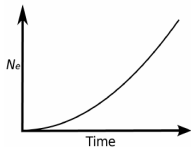
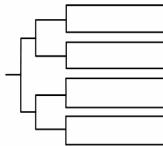
Phylogenetics of fast-evolving viruses

Inferring spatial and temporal dynamics from genomic data:

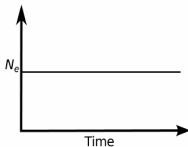
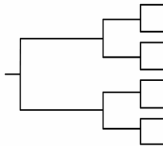
Phylogenies*!

* plus complicated models

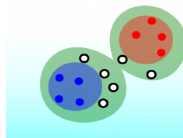
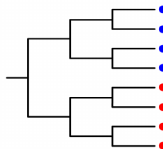
A
Exponential Growth



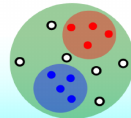
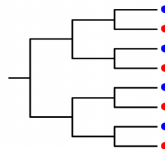
B
Constant Population Size



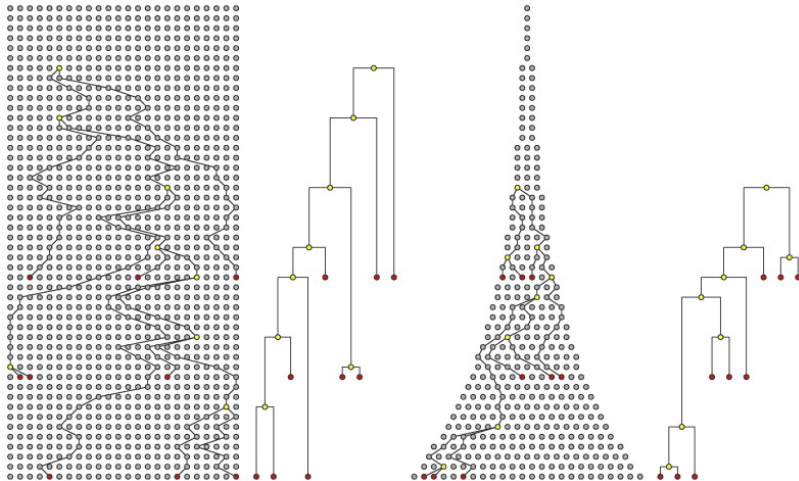
A
Structured Host Population



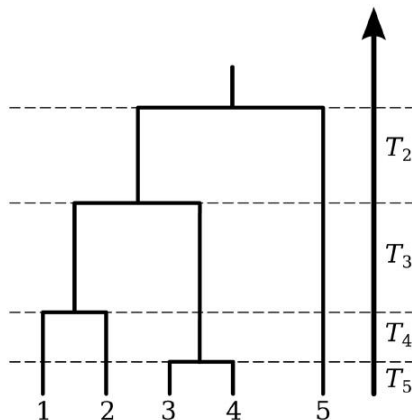
B
Unstructured Host Population



Trees and the coalescent



Central object: time-calibrated trees



Let T_n denote the time for n lineages to *coalesce*, i.e., merge into one ancestral lineage, in a population of size N_e . Then:

$$\Pr(T_n = t) = \lambda_n e^{-\lambda_n t}$$

$$\lambda_n = \binom{n}{2} \frac{1}{N_e} = \binom{n}{2} \frac{1}{N_e \tau}$$

where N_e is the effective population size and τ is the generation time. Let T_{mrca} denote the age of the most recent common ancestor:

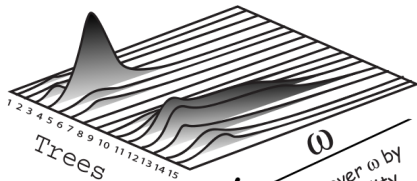
$$\begin{aligned} \mathbb{E}[T_{\text{mrca}}] &= \mathbb{E}[T_n] + \mathbb{E}[T_{n-1}] + \dots + \mathbb{E}[T_2] \\ &= 1/\lambda_n + 1/\lambda_{n-1} + \dots + 1/\lambda_2 \\ &= 2N_e \left(1 - \frac{1}{n}\right) \end{aligned}$$

Figure: Figure 4 from [Volz et al. \(2013\)](#).

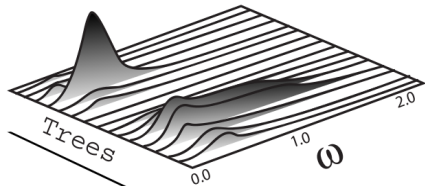
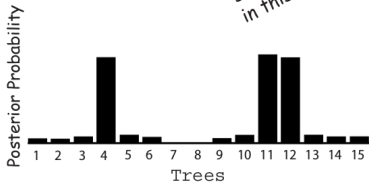
$$p(t, \mathbf{b}, \boldsymbol{\omega} | D) = \frac{f(D | t, \mathbf{b}, \boldsymbol{\omega}) \pi(t, \mathbf{b}, \boldsymbol{\omega})}{\sum_{t_i \in T_n} \int_B \int_{\Omega} f(D | t_i, \mathbf{b}_i, \boldsymbol{\omega}) \pi(t_i, \mathbf{b}_i, \boldsymbol{\omega}) d\boldsymbol{\omega} d\mathbf{b}_i} \quad (1)$$

- ⊙ D : observed sequence (DNA) data;
- ⊙ T_n : set of all binary ranked trees;
- ⊙ $\mathbf{b}_k$: set of branch lengths of $t_k \in T_n$ ($\mathbb{R}_+^{2n-2}$, kind of) ;
- ⊙ $\boldsymbol{\omega}$: set of parameters of interest such as substitution model parameters, migration rates, heritability coefficients, etc.

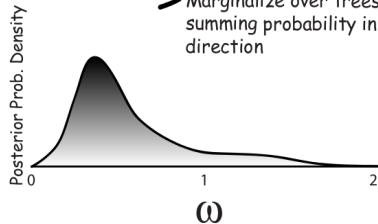
The end product



← Marginalize over ω by summing probability in this direction

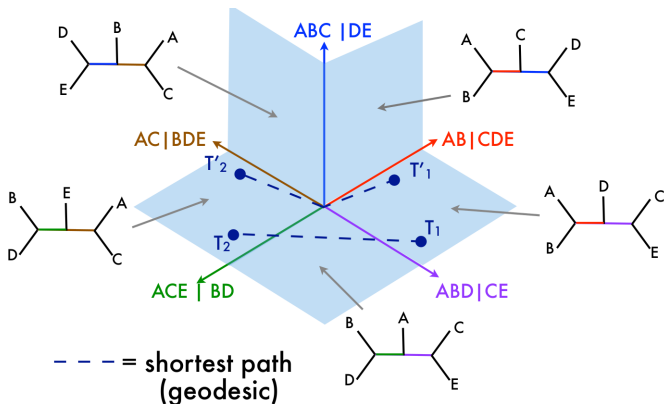


→ Marginalize over trees by summing probability in this direction



This place is weird..

Traversing cubic complexes efficiently²



Applications: Molecular Epidemiology, Evolutionary Biology.

²<https://youtu.be/h9bWRQ6aeKA>

(Adaptive) Metropolis-Hastings for trees

General MH setup.

Let $\tau = (t, \mathbf{b})$ denote a tree with topology t and branch lengths $\mathbf{b}$. For two trees τ and τ' , denote the transition kernel by $q_\gamma(\tau|\tau') := \Pr(\tau' \rightarrow \tau|\gamma)$.

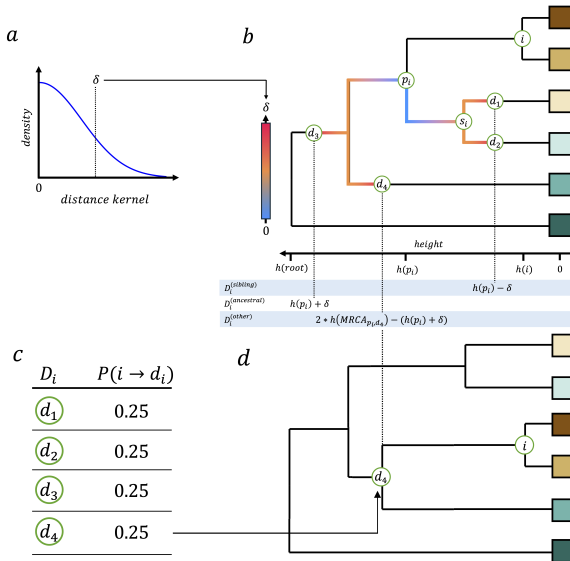
Accepting with probability

$$A_\gamma(\tau|\tau') = \min \left(1, \frac{p(\tau', \omega|D)q_\gamma(\tau|\tau')}{p(\tau, \omega|D)q_\gamma(\tau'|\tau)} \right)$$

leads to the desired target.

Note: Here $\gamma > 0$ is a so-called tuning parameter.

SubTreeLeap



STL – ergodicity

See [Brusselmans et al. \(2026\)](#) and [Carvalho \(2019\)](#), Chapter 2.

Lemma

Assume strictly positive branch lengths. Then SubTreeLeap induces an irreducible Markov chain on T_n .

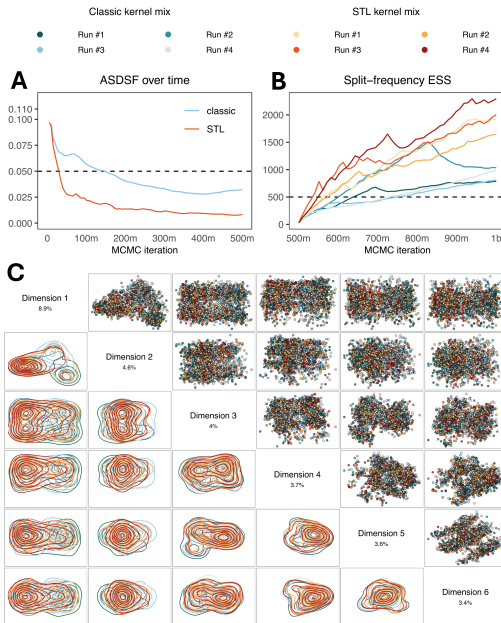
Sketch: Starting at $x \in T_n$, notice there exists $\delta_y^\star > 0$ such that $P(x \rightarrow y \mid \delta_y^\star) > 0$ for any tree $y \in T_n$ in the SPR neighbourhood of x .

Theorem

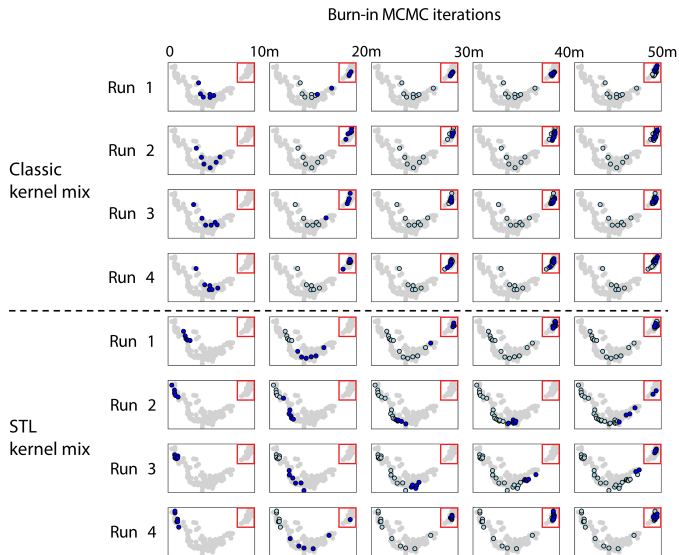
Assume the target satisfies $p(A) > 0$ for all $A \in \Psi$. Then, SubTreeLeap induces an ergodic Markov chain on Ψ .

Sketch: Employ the lemma to get to the case where $d_{\text{SPR}}(x, y) = 0$ and then establish Harris recurrence.

Convergence and posterior visualisation

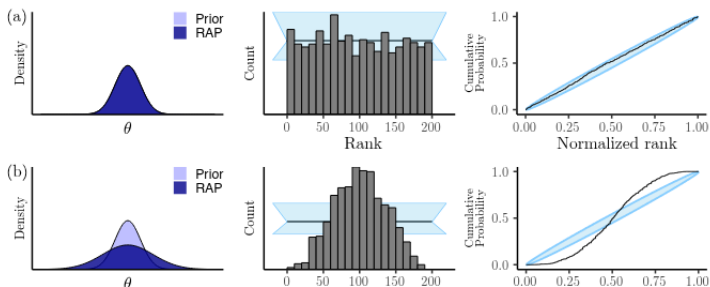


Burning in



How do you tell if it worked, though?

- ⊙ **Super important** to check consider actual tree metrics when checking convergence (convergence on ω is not informative)³.
- ⊙ **Validating** the methodology using rigorous modern techniques such as Simulation-based Calibration (SBC)⁴



³Brusselmans et al (2024) *Virus Evolution*

⁴Mendes et al. (2025) *Systematic Biology*

Open problems in MCMC for phylogenies

Open problems:

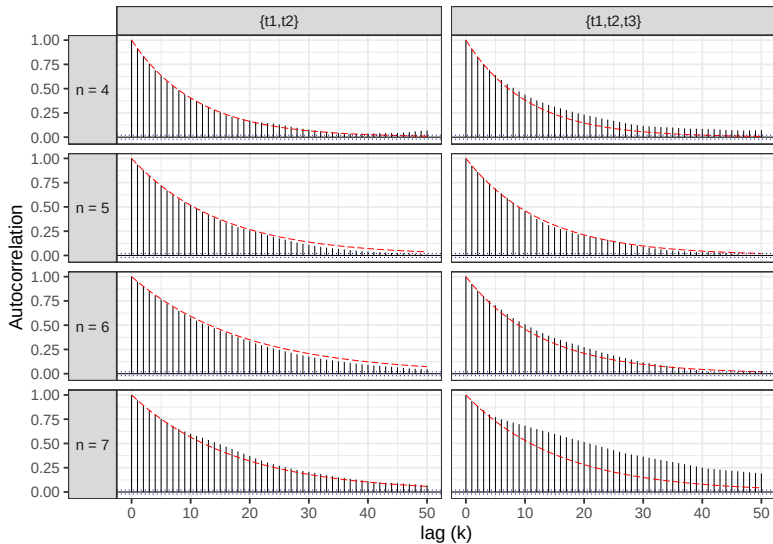
- How can we construct (much) more efficient proposals?
How to exploit structure? **Geometry!**
- Optimal scaling: what's the **optimal** acceptance probability?
- New, statistically-motivated metrics and measures of centrality (e.g. HIPSTR⁵.)
- (*) How to quantify exploration of the target? (Custom) **Tools!**
- (*) Theoretical aspects of phylogenetic MCMC⁶;

⁵Baele et al (2025) *Bioinformatics*.

⁶See Semper & Palacios (2022) and Alves, Saporito & Carvalho (2024) for recent attempts.

My lovely tree lumps... Check it out!

Alves, Saporito & Carvalho (2024)



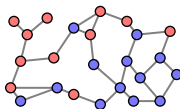
Teste de Duas Amostras

- Sejam $X_1, \dots, X_n \sim F_1$ e $Y_1, \dots, Y_n \sim F_2$.

- Considere o teste de hipóteses:

$$H_0 : F_1 = F_2 \quad \text{vs} \quad H_1 : F_1 \neq F_2$$

- Seja G_n o grafo com vértices $\{X_1, \dots, X_n\} \cup \{Y_1, \dots, Y_n\}$.
- Vértices em G_n são conectados com base em sua proximidade.



- $N(G_n) :=$ Número de arestas monocromáticas.
- Teste: **rejeite H_0 se $N(G_n)$ é grande**
- $G_n =$ Minimum Spanning Tree (Friedman and Rafsky 1979).
- $G_n =$ k-Nearest Neighbour Graph (Schilling 1986; Henze 1988).
- $G_n =$ Minimum Distance Pairing (Rosenbaum 2005).
- Diferentes **opções de métricas** $\Rightarrow$ diferentes opções para G_n .



A ESCOLA

Excelência na aplicação da matemática e da computação

Com método de ensino participativo e professores altamente qualificados, a FGV EMap prepara profissionais para desenvolverem matemática e modelos computacionais de forma inovadora, adaptados aos desafios da era da informação.

Aqui, os alunos saem com uma base sólida para atuar em setores estratégicos de organizações públicas e privadas, qualificando-os, também, para a realização de pesquisas acadêmicas e projetos de consultoria.

Conheça a escola



<https://emap.fgv.br/>



- Mestrado e Doutorado em **Matemática Aplicada e Ciência de Dados**;
- Edital aberto até 02/10 (**corram!**);
- Seleção: análise de currículo, e se aprovado, entrevista;
- **Bolsa de mestrado: R\$ 3.000,00;**
- **Bolsa de doutorado: R\$ 4.000,00;**
- Possibilidade de doutorado direto;
- Bolsas de projetos e monitoria.

Mestrado

- 7 disciplinas: 4 centrais e 3 eletivas
- 2 disciplinas de seminários

Doutorado

- 7 disciplinas: 3 centrais e 4 eletivas
- 4 disciplinas de seminários
- Exame de Qualificação I, II, III e Tese

■ Centrais do Doutorado (e Mestrado)

Medida, Integração e Probabilidade	Otimização
Teoria Qualitativa de EDO	Controle Ótimo
Análise Numérica e Simulação	Processos Estocásticos
Análise Funcional: Fundamentos	Equações Diferenciais Estocásticas: Teoria e Simulação
Análise Funcional: Operadores Lineares	Finanças Quantitativas
Ciência de Redes	Métodos Quantitativos em Risco
Equações Diferenciais Parciais e Aplicações	Biologia Matemática
Visualização para Aprendizado de Máquinas	Causalidade
Aprendizado de Máquinas	Epidemiologia Matemática
Modelos Generativos	Inferência Estatística
Redes Neurais e Aprendizado Profundo	Estatística Computacional
Aprendizado por Reforço	Estatística Bayesiana

Nossos egressos



Lucas Moschen
Imperial College
London (antes,
Sorbonne)



Lucas Resck
Cambridge
University



Vitória Guardieiro
University of
Pennsylvania

Take home

Trees are cool, useful, ...

... and hard! A worthy statistical opponent.

Computational methods are key

Learn to program and learn Computational Statistics⁷

We've got loads to do!

Biomedical statistics is where most of the cool data and problems **with actual impact** are.

Come work with us!

Come to the School of Applied Mathematics (FGV EMaP)

⁷Here's a place to start:

https://github.com/maxbiostat/Computational_Statistics

THE
END